WIEFERICH PRIMES IN ORDOWSKI'S CONGRUENCES

DEEP

Abstract. We prove that no odd number with a repeated prime factor satisfies Ordowski's congruence for every base, where the congruence asks that a base raised to the number and raised to its largest prime exponent leave the same residue modulo the number. For the base two such numbers do exist. Every prime carrying the largest exponent of one of them is a Wieferich prime, and the smallest example is divisible by the square of 3511. Both answers are verified in Lean.

1. Introduction

For a positive integer n let $a(n)$ be the largest exponent in the prime factorization of n, so that $a(1) = 0$, $a(12) = 2$ and $a(n) = 1$ exactly when $n > 1$ is squarefree. This is sequence A051903 of the OEIS [11]. On 2 December 2019 Thomas Ordowski recorded there three questions about this function.

(Q1) Is there a composite $n > 4$ with $n \equiv a(n) \pmod{\varphi(n)}$?
(Q2) Is there an odd n with $a(n) > 1$ and $n \equiv a(n) \pmod{\lambda(n)}$?
(Q3) Is there an odd n with $a(n) > 1$ and $2^n \equiv 2^{a(n)} \pmod{n}$?

Here φ is Euler's function and λ is Carmichael's function [1]. Ordowski added that a negative answer to (Q3) would give a negative answer to (Q2). The first question is equivalent to Lehmer's totient problem [7], an observation recorded in the Formal Conjectures project [4], where all three questions are stated in Lean and marked as open.[1]

We settle (Q2) and (Q3). Their answers differ, so the reduction noted by Ordowski does not apply.

Theorem 1. *No odd n with $a(n) \geq 2$ satisfies $b^n \equiv b^{a(n)} \pmod{n}$ for every integer b. More precisely, if p is a prime with $p^{a(n)} \mid n$, then the base $b = p + 1$ fails.*

Theorem 2. *The number*

$$n_0 \;=\; 7 \cdot 631 \cdot 881 \cdot 3511^2 \cdot 201961 \;=\; 9687963167864344937$$

satisfies $a(n_0) = 2$ and $2^{n_0} \equiv 2^2 \pmod{n_0}$. It is the least odd n with $a(n) > 1$ and $2^n \equiv 2^{a(n)} \pmod{n}$.

Both theorems come from one inequality. If p is an odd prime that attains the largest exponent E of n and $b^n \equiv b^E \pmod{p^E}$, then lifting the exponent gives $E \leq v_p(b^{p-1} - 1) + v_p(E)$ (Proposition 4). Two consequences follow. The prime p must be a base-b Wieferich

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[1] They appear there as `OeisA51903.conjecture1`, `conjecture2` and `conjecture3`. The second is stated in the equivalent form $b^n \equiv b^{a(n)} \pmod{n}$ for every b. The two forms agree because $a(n)$ is at least every exponent of n, so both sides vanish modulo $p^{v_p(n)}$ whenever p divides b, while for b prime to n the condition is $\lambda(n) \mid n - a(n)$.

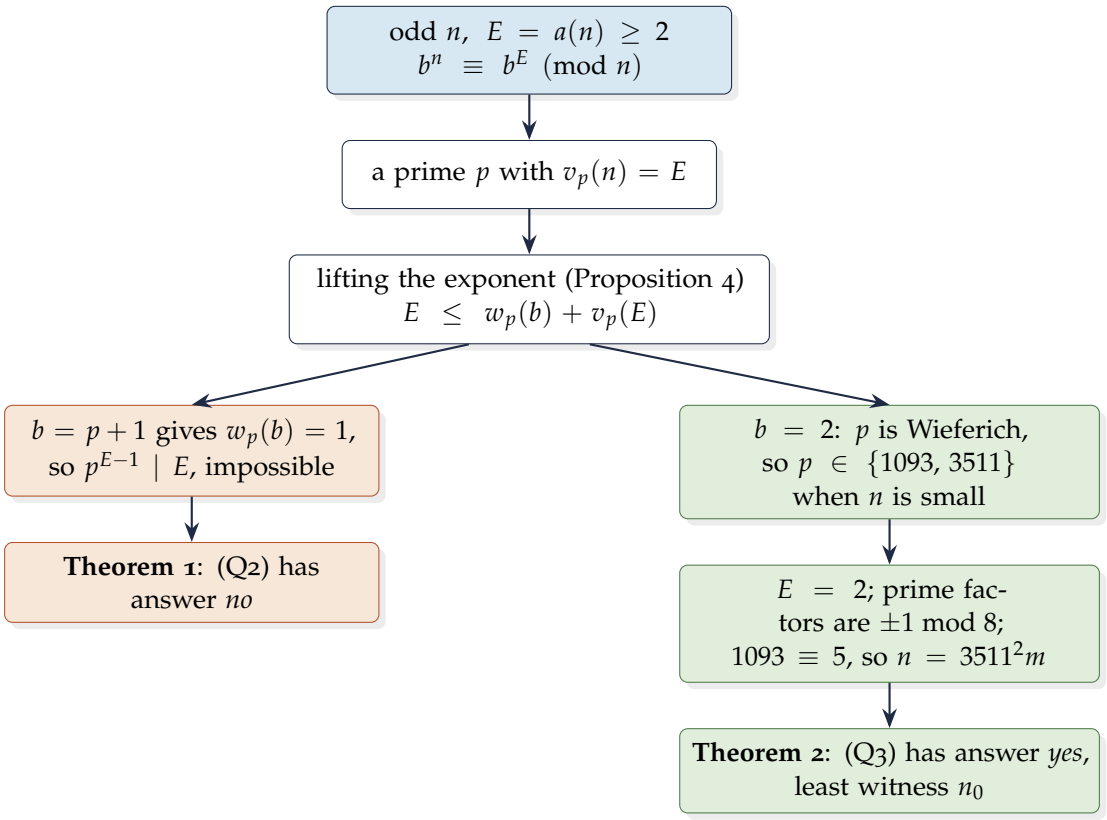


FIGURE 1. The structure of the argument. The left branch uses one base for each prime; the right branch fixes the base two.

prime, which no prime is for the base $p+1$; this gives Theorem 1. For the base two it forces the square of 1093 or 3511 into every small witness, and a quadratic residue argument removes 1093. What remains is a search through numbers of the form $3511^2 m$, which produces n_0. Figure 1 shows the route.

Witnesses for the base two are plentiful once one is known: each one produces another from a primitive prime divisor of a Mersenne number (Corollary 8), and there are exactly 28 of them up to $3.49 \cdot 10^{20}$ (Theorem 9).

Everything in Theorem 1 and the first half of Theorem 2, together with the structural results of section 4, has been verified in the Lean proof assistant against the statements of the Formal Conjectures project. The minimality claim rests on a computation that we describe in section 5.

2. Lifting the exponent

For a prime p and a nonzero integer x we write $v_p(x)$ for the exponent of p in x. For an odd prime p and an integer $b \geq 2$ prime to p we put

$$w_p(b) = v_p\big(b^{p-1} - 1\big),$$

which is at least 1 by Fermat's theorem. The prime p is a *base-b Wieferich prime* if $w_p(b) \geq 2$. For $b = 2$ these are the classical Wieferich primes [13]; only two are known, 1093 and 3511,

and there are no others below $6.7 \cdot 10^{15}$ [3].[2] We write $w_p = w_p(2)$; a direct computation gives $w_{1093} = w_{3511} = 2$.

The following lemma is the classical lifting of the exponent; we include the short proof to keep the paper self-contained.

Lemma 3. *Let p be an odd prime, let $x \geq 2$ be an integer with $x \equiv 1 \pmod{p}$, and let $k \geq 1$. Then $v_p(x^k - 1) = v_p(x-1) + v_p(k)$.*

Proof. If $p \nmid t$ then $(x^t - 1)/(x-1) = 1 + x + \cdots + x^{t-1} \equiv t \not\equiv 0 \pmod{p}$, so $v_p(x^t - 1) = v_p(x-1)$. If $y = 1 + p^r u$ with $r \geq 1$ and $p \nmid u$, the binomial theorem gives

$$y^p = 1 + p^{r+1}u + \binom{p}{2} p^{2r} u^2 + \sum_{j \geq 3} \binom{p}{j} p^{jr} u^j,$$

and since p is odd, $p \mid \binom{p}{2}$, so every term after the second is divisible by p^{r+2}. Hence $v_p(y^p - 1) = v_p(y-1) + 1$. Writing $k = p^s t$ with $p \nmid t$ and applying the second fact s times to $y = x^t$ proves the lemma. □

Proposition 4. *Let p be an odd prime, let $b \geq 2$ be prime to p, and let $n \geq 1$ and $E \geq 1$ satisfy $p^E \mid n$ and $b^n \equiv b^E \pmod{p^E}$. Then*

$$E \;\leq\; w_p(b) + v_p(n - E) \;=\; w_p(b) + v_p(E).$$

Proof. Since $n \geq p^E > E$, the integer $n - E$ is positive, and $v_p(E) < E \leq v_p(n)$ gives $v_p(n-E) = v_p(E)$. As b is prime to p we may cancel b^E and obtain $b^{n-E} \equiv 1 \pmod{p^E}$. Raising to the power $p-1$ gives $x^{n-E} \equiv 1 \pmod{p^E}$ with $x = b^{p-1} \equiv 1 \pmod{p}$, and Lemma 3 yields

$$E \;\leq\; v_p\big(x^{n-E} - 1\big) \;=\; w_p(b) + v_p(n-E). \qquad \square$$

Corollary 5. *Under the hypotheses of Proposition 4 with $E \geq 2$, the prime p is a base-b Wieferich prime.*

Proof. If $w_p(b) = 1$, then $v_p(E) \geq E - 1$ by Proposition 4, so $p^{E-1} \mid E$. But $p^{E-1} \geq 3^{E-1} \geq 2E - 1 > E$ for $E \geq 2$. □

Figure 2 pictures the mechanism. The units $x \equiv 1 \pmod{p}$ form a descending chain of layers according to $v_p(x-1)$, and raising to the power k moves a unit up exactly $v_p(k)$ layers. A witness must push $x = b^{p-1}$ to layer E, while the exponent $n - E$ contributes exactly $v_p(E)$ layers.

3. Every base at once

Proof of Theorem 1. Let n be odd with $E = a(n) \geq 2$, and choose a prime p with $v_p(n) = E$. It is odd because n is. Put $b = p + 1$, which is prime to p. By Lemma 3 applied to $x = p + 1$ and $k = p - 1$,

$$w_p(p+1) = v_p\big((p+1)^{p-1} - 1\big) = v_p(p) + v_p(p-1) = 1.$$

So p is not a base-$(p+1)$ Wieferich prime. If $b^n \equiv b^E \pmod{n}$, the same holds modulo p^E, and Corollary 5 gives a contradiction. □

[2] Earlier searches reached $6 \cdot 10^9$ (Lehmer [8]), $4 \cdot 10^{12}$ (Crandall, Dilcher and Pomerance [2]) and $1.25 \cdot 10^{15}$ (Knauer and Richstein [6]).

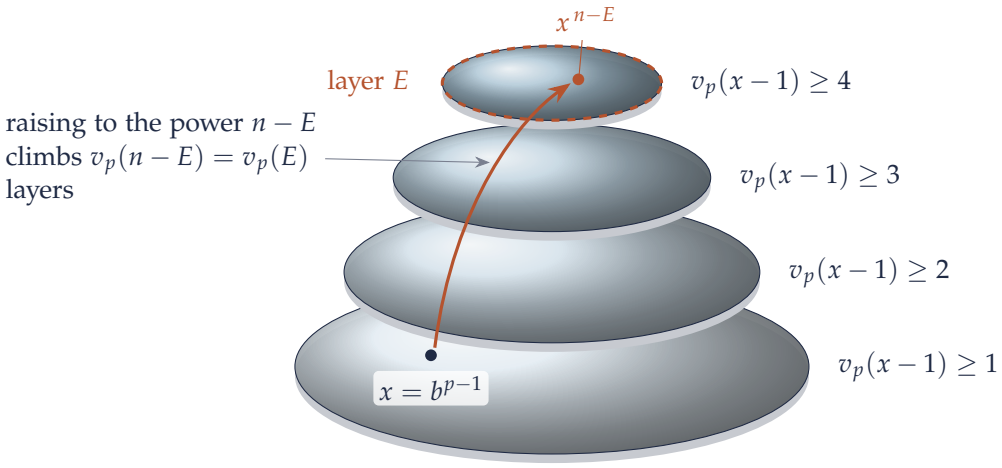


FIGURE 2. The layers $\{x \equiv 1 \pmod{p^k}\}$ drawn as stacked discs. A unit on layer $w_p(b)$ reaches layer E only if $E \leq w_p(b) + v_p(E)$; for $E \geq 2$ this needs $w_p(b) \geq 2$.

Remark 6. In the language of (Q2) the proof says that p^{E-1}, which divides $\lambda(n)$, cannot divide $n - E$. This is the same mechanism that reduces (Q1) to Lehmer's problem in [4]. The point of Proposition 4 is that it records what a single base forces, and this is what the question for the base two requires.

4. The base two

From now on a *witness* is an odd n with $a(n) \geq 2$ and $2^n \equiv 2^{a(n)} \pmod{n}$.

Theorem 7. *Let n be a witness and $E = a(n)$.*

(1) *Every prime p with $v_p(n) = E$ is a Wieferich prime, and $E \leq w_p + v_p(E)$.*
(2) *If E is even, every prime factor q of n satisfies $q \equiv \pm 1 \pmod 8$.*
(3) *If every prime p with $v_p(n) = E$ is below $6.7 \cdot 10^{15}$, which holds in particular when $n < 4.4 \cdot 10^{31}$, then $E = 2$ and*

$$n = 3511^2 m, \qquad m \text{ squarefree}, \qquad 3511 \nmid m, \qquad m \equiv 1757 \pmod{3510}.$$

Proof. Part (1) is Proposition 4 and Corollary 5 with $b = 2$.

For (2), let q be a prime factor of n; it is odd. Cancelling 2^E gives $2^{n-E} \equiv 1 \pmod q$, and $n - E = 2k + 1$ is odd. Hence $2 \equiv 2^{2k+2} = (2^{k+1})^2 \pmod q$, so 2 is a quadratic residue modulo q, which happens exactly when $q \equiv \pm 1 \pmod 8$ [5, Ch. 5].

For (3), each prime p with $v_p(n) = E$ is a Wieferich prime below $6.7 \cdot 10^{15}$ by (1), so $p \in \{1093, 3511\}$ by [3], and $w_p = 2$. If $E \geq 3$ then $p^{E-2} \mid E$ by (1), which is impossible since $p^{E-2} \geq 1093^{E-2} > E$. Thus $E = 2$. Now every prime whose square divides n attains the exponent E, so it is 1093 or 3511. Part (2) applies because E is even, and $1093 \equiv 5 \pmod 8$, so $1093 \nmid n$. Hence $n = 3511^2 m$ with m squarefree and prime to 3511. The order of 2 modulo 3511^2 is $1755 = 3^3 \cdot 5 \cdot 13$, the same as modulo 3511; one checks that $2^{1755} \equiv 1 \pmod{3511^2}$ and that $2^{1755/r} \not\equiv 1 \pmod{3511}$ for $r = 3, 5, 13$. From $2^{n-2} \equiv 1 \pmod{3511^2}$ we get $1755 \mid 3511^2 m - 2$, and since $3511 \equiv 1 \pmod{1755}$ this says $m \equiv 2 \pmod{1755}$. As m is odd, $m \equiv 1757 \pmod{3510}$. □

TABLE 1. The prime power factors of n_0. Each order divides $19305 = 3^3 \cdot 5 \cdot 11 \cdot 13$, and $n_0 \equiv 2 \pmod{19305}$.

q^e	q mod 8	order of 2 modulo q^e	where q comes from
7	7	3	$2^3 - 1$
631	7	$45 = 3^2 \cdot 5$	$2^{45} - 1$
881	1	$55 = 5 \cdot 11$	$2^{55} - 1$
3511^2	7	$1755 = 3^3 \cdot 5 \cdot 13$	Wieferich
201961	1	$55 = 5 \cdot 11$	$2^{55} - 1$

Proof that n_0 is a witness. The factorization of n_0 shows $a(n_0) = 2$. For each prime power $q^e \parallel n_0$, Table 1 lists the order of 2 modulo q^e. All of these orders divide 19305, and $n_0 - 2 = 19305 \cdot 501836993932367$, so $2^{n_0-2} \equiv 1$ modulo every q^e and hence modulo n_0. Multiplying by 4 gives $2^{n_0} \equiv 2^2 \pmod{n_0}$. □

The table explains how n_0 was found. Once 3511^2 is fixed, it is natural to look for m among products of primes q whose order $\mathrm{ord}_q(2)$ divides a small multiple L of 1755. Then the only remaining condition is $3511^2 m \equiv 2 \pmod{L}$, a congruence on a product of residues. With $L = 19305$ the primes dividing $2^d - 1$ for $d \mid L$ already contain the solution above.[3]

Corollary 8. *If n is a witness with $a(n) = 2$ and q is a prime with $\mathrm{ord}_q(2) = n - 2$, then nq is a witness with $a(nq) = 2$. Consequently there are infinitely many witnesses.*

Proof. Since $n - 2 = \mathrm{ord}_q(2)$ divides $q - 1$, and $q \neq n - 1$ because $n - 1$ is even, we have $q \geq 2(n-2) + 1 > n$. So q is odd and $q \nmid n$; thus nq is odd and $a(nq) = 2$. Modulo q, the exponent $nq - 2 = (n-2)q + 2(q-1)$ is a multiple of $n - 2$. Modulo n, the order of 2 divides $n - 2$, which divides $q - 1$, and $nq - 2 = n(q-1) + (n-2)$. Hence $2^{nq-2} \equiv 1 \pmod{nq}$. A prime q with $\mathrm{ord}_q(2) = n - 2$ exists by Zsigmondy's theorem [14], so starting from n_0 the construction never stops. □

5. The least witness

Proof that n_0 is the least witness. Let $n \leq n_0$ be a witness. Every prime p with $v_p(n) = a(n)$ satisfies $p^2 \leq n$, so $p < 3.12 \cdot 10^9$. We tested every prime below $4.2 \cdot 10^9$ and found that 1093 and 3511 are the only Wieferich primes in this range, in agreement with [8, 2]. The argument for part (3) of Theorem 7 then applies verbatim, so $n = 3511^2 m$ with

$$m \equiv 1757 \pmod{3510}, \qquad m \equiv \pm 1 \pmod{8}, \qquad 3511 \nmid m, \qquad 2^{3511^2 m - 2} \equiv 1 \pmod{m}.$$

The second condition comes from part (2) of Theorem 7, and the last is the witness condition modulo m. We enumerated all $m \leq 785906390297$ in these two residue classes modulo 14040, about $1.12 \cdot 10^8$ values, and computed $2^{3511^2 m - 2} \bmod m$ with Montgomery multiplication. The first value satisfying the congruence conditions is $m = 785906390297 = 7 \cdot 631 \cdot 881 \cdot 201961$, which is squarefree and gives $n = n_0$. □

The same search, run up to $m \leq 28281899434817$, finds every witness up to

$$N_1 = 3511^2 \cdot 28281899434817 = 348634396442820771857 \approx 3.49 \cdot 10^{20}.$$

[3]The first witness we found this way was $3511^2 \cdot 7 \cdot 31 \cdot 73 \cdot 79 \cdot 89 \cdot 271 \cdot 937 = 34863436442820771857$.

TABLE 2. All witnesses $n = 3511^2 m \le 348634396442820771857$, with the prime factors of m.

	n	m
1	9687963167864344937	$7 \cdot 631 \cdot 881 \cdot 201961$
2	10191389002728947327	$439 \cdot 631 \cdot 1151 \cdot 2593$
3	11613371395592377127	$23 \cdot 73 \cdot 89 \cdot 1327 \cdot 4751$
4	12853981953529455377	$71 \cdot 73 \cdot 79 \cdot 751 \cdot 3391$
5	12874928605806902897	$31 \cdot 79 \cdot 151 \cdot 431 \cdot 6553$
6	13730896205770900457	$7 \cdot 31 \cdot 271 \cdot 18941231$
7	15996167795452598327	$7 \cdot 79 \cdot 23279 \cdot 100801$
8	18091379716837511177	$127 \cdot 20231 \cdot 571201$
9	21095218232071695857	$23 \cdot 151 \cdot 6553 \cdot 75193$
10	29350506332796150647	$73 \cdot 79 \cdot 89 \cdot 199 \cdot 23311$
11	39992485447538478857	$71 \cdot 199 \cdot 263 \cdot 881 \cdot 991$
12	67350552227362140167	$73 \cdot 191 \cdot 199 \cdot 1969111$
13	73657883659512951047	$311 \cdot 16183 \cdot 1187239$
14	82340163204173730767	$79 \cdot 36793 \cdot 2298041$
15	98848358486672333177	$71 \cdot 73 \cdot 151 \cdot 1801 \cdot 5689$
16	139478761055861260607	$7 \cdot 89 \cdot 151 \cdot 991 \cdot 121369$
17	161001183676264892417	$47 \cdot 71 \cdot 487 \cdot 599 \cdot 13417$
18	164739335550634207937	$7 \cdot 89 \cdot 1567 \cdot 2089 \cdot 6553$
19	191711593813855690577	$79 \cdot 4201 \cdot 6553 \cdot 7151$
20	201401736552101605457	$7 \cdot 103 \cdot 311 \cdot 6553 \cdot 11119$
21	204945450678447017177	$45137 \cdot 368335801$
22	223934360665868890247	$7 \cdot 31 \cdot 66161 \cdot 1265311$
23	225021378651588304697	$73 \cdot 79 \cdot 151 \cdot 271 \cdot 77351$
24	259428933707962157687	$71 \cdot 911 \cdot 2647 \cdot 122921$
25	275690737952807302127	$71 \cdot 751 \cdot 881 \cdot 476087$
26	316763115741760627487	$31 \cdot 631 \cdot 10687 \cdot 122921$
27	324284460815828474327	$271 \cdot 337 \cdot 11471 \cdot 25111$
28	348634396442820771857	$7 \cdot 31 \cdot 73 \cdot 79 \cdot 89 \cdot 271 \cdot 937$

Theorem 9. *There are exactly* 28 *witnesses* $n \le N_1$. *All of them have* $a(n) = 2$ *and the form* $3511^2 m$ *with* m *squarefree; they are listed in Table 2.*

Proof. For a witness $n \le N_1$, the primes attaining $a(n)$ lie below $\sqrt{N_1} < 1.9 \cdot 10^{10}$, a range in which 1093 and 3511 are the only Wieferich primes [2]. As in the previous proof, $n = 3511^2 m$ with m in the two residue classes modulo 14040 and $m \le 28281899434817$. All $4.03 \cdot 10^9$ such m were tested. The 28 values that satisfy the congruence were then checked again in exact integer arithmetic; each is squarefree, prime to 3511, and gives $2^n \equiv 4 \pmod{n}$. □

The largest entry of Table 2 is the first witness we found by the construction above. Figure 3 shows how the witnesses spread out.

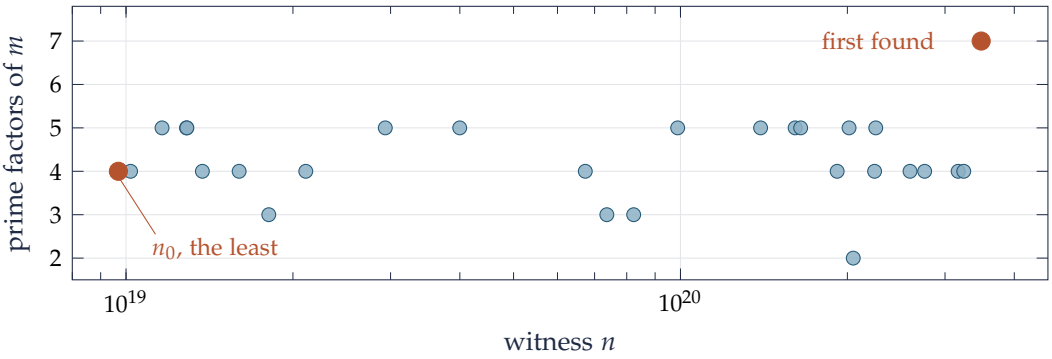


FIGURE 3. The 28 witnesses up to N_1, by size and by the number of prime factors of m. Every one of them is divisible by 3511^2.

6. Formal verification

The file `ErdosProofs/Ordowski.lean` in the accompanying repository [12] proves the following in Lean 4 [10] with Mathlib [9].[4]

- `conjecture2`: the statement of (Q2) from [4] with answer `False` (Theorem 1).
- `conjecture3`: the statement of (Q3) with answer `True`, using the witness n_0.
- `le_padicValNat_of_modEq` and `wieferich_of_modEq`: the inequality $E \le w_p(b) + v_p(n - E)$ of Proposition 4 and Corollary 5, for every base.
- `base_two_wieferich` and `prime_factor_mod_eight`: parts (1) and (2) of Theorem 7.

Lifting the exponent is Mathlib's `padicValNat.pow_sub_pow`, and the quadratic character of 2 is `ZMod.exists_sq_eq_two_iff`. The congruence $2^{n_0} \equiv 4 \pmod{n_0}$ cannot be checked by expanding 2^{n_0}, which has about 10^{19} binary digits. Instead the file defines binary modular exponentiation, proves that it computes $b^e \bmod m$, and lets the kernel evaluate it in sixty-four squaring steps. The factorization of n_0 is certified by Mathlib's `norm_num` primality extension. No step uses `native_decide`, so the proofs rely only on the Lean kernel and the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The search behind section 5 is not formalized; its C source is in the repository.

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